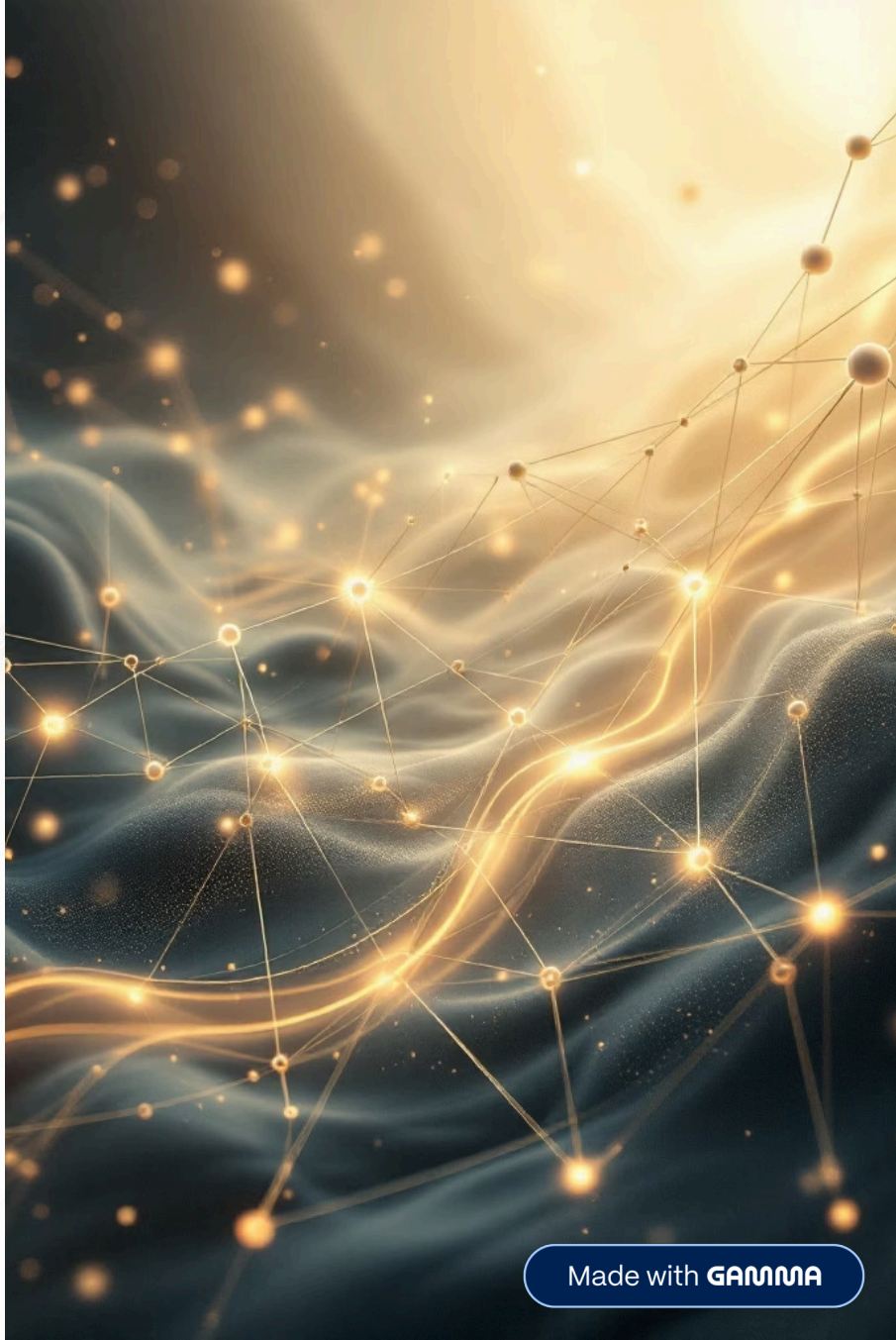
Customer Churn Prediction

Machine Learning for Customer Retention



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Project Objective

Predict Churn

Identify customers likely to leave

Best Model

Compare and identify top performer

Business Decisions

Support proactive retention strategies

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Dataset Overview

File: Cleaned_Customer_Churn.csv

Target Variable: Churn

Features Include:

- Customer demographics
- Account details
- Service usage patterns
- Subscription information

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Technologies & Tools

Python

Core programming language

Pandas & NumPy

Data manipulation and numerical operations

Matplotlib

Data visualization

Scikit-learn

ML models and evaluation

XGBoost

Advanced gradient boosting

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Project Workflow

Data Loading

Import dataset using Pandas, verify structure and cleaned data

Data Preparation

Split into features (X) and target (y). Apply train-test split (80% training, 20% testing)

Model Building

Implement multiple classification algorithms

Evaluation

Assess performance using metrics

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Models Implemented

1

Logistic Regression

Baseline model for binary classification

2

Decision Tree Classifier

Tree-based model for interpretability

3

Random Forest Classifier

Ensemble model improving accuracy and reducing overfitting

4

XGBoost Classifier

High-performance gradient boosting model

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Evaluation Metrics

Accuracy Score

Classification Report

- Precision
- Recall
- F1-score

Confusion Matrix

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Model Comparison

Logistic Regression

Simple and interpretable baseline model

Decision Tree

Rule-based model, easy to visualize

Random Forest

Ensemble model with improved accuracy

XGBoost

Advanced boosting with high performance

Key Insight: Ensemble models like Random Forest and XGBoost generally perform better. All model results combined into DataFrame for easy comparison.

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Business Impact

Reduce Customer Loss

Proactive retention strategies

Increase Revenue

Improve customer satisfaction

Data-Driven Decisions

Support strategic planning

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Future Enhancements

Hyperparameter Tuning

GridSearchCV optimization

Feature Analysis

Importance analysis

Real-Time System

Streamlit/Flask deployment

Interactive Dashboard

Power BI/Tableau

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